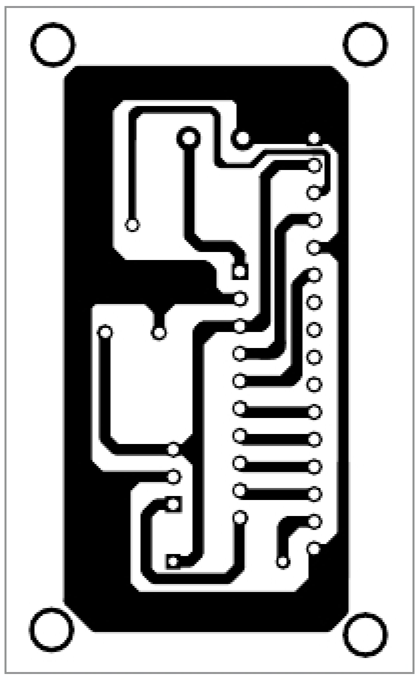

Fig. 3: Actual-size PCB layout of the noise level monitor

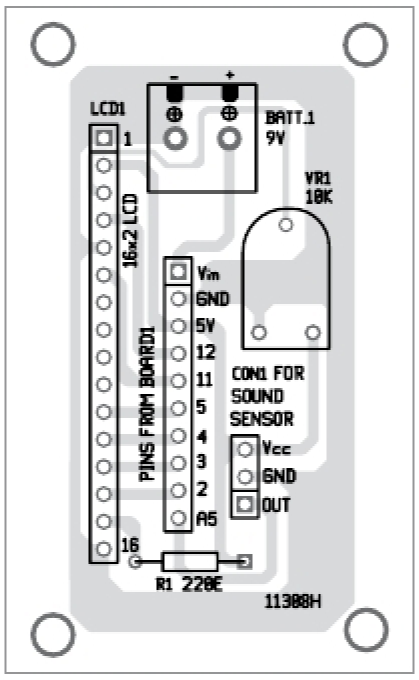

Fig. 4: Components layout for the PCB

level monitor is shown in Fig. 3 and its components layout in Fig. 4. Assemble the components on the PCB as per the circuit diagram. In this project, analogue OUT pin of the sound sensor is connected to A5 pin of Arduino, Vcc pin to 5V and GND pin to ground pin (GND) of Arduino.

Connect Arduino to a computer using a USB cable. Open the source code from Arduino IDE, select the proper COM port and board from Tools menu. After selecting the proper COM port and board, compile and upload the source code to the board.

After uploading the code, remove the USB cable and connect a 9V battery as power supply for standalone use. After 9V battery is connected to the circuit, you will see Audio Meter on LCD1. Now, speak in front of the microphone. The sound detected will be converted to corresponding analogue voltage, digital voltage and sound level in dB, and all these values will be displayed on LCD1. You can use this circuit to capture the noise around the meter.

The circuit has been tested for various situations and works fine. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*

PARTS LIST	
Semiconductors:	
Board1	- Arduino Uno board
	- Sound sensor module
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 220-ohm
VR1	- 10-kilo-ohm preset
Miscellaneous:	
BATT.1	- 9V battery
LCD1	- 16x2 alphanumeric LCD
CON1	- 3-pin connector for sound sensor module
	- 10-pin connector for Board1

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